

OPERSTACK

Sixteen gates

The checks a content site must pass before it ships, in the order they run, with the failure each one was written for and the fix. Free, open source, one command.

`npx @operstack/gates`

1,660

MDX files in one real run

13 s

for all sixteen gates on a laptop

11

content sites the gates were built on

Every gate in this document exists because that exact defect shipped at least once and the build log said nothing. The examples come from a fictional demo site, Isla Verde, that ships with the package. Every number in the examples is invented.

What ships when nobody checks

Every content site that grows past a few hundred pages accumulates the same debt. None of it fails a build. All of it fails a reader, and answer engines stop citing a site long before a person complains.

- 01 A currency sign that renders as a question mark in the search snippet.**
Pasted from a spreadsheet. Fine in the editor, wrong on the results page.

- 02 A title cut on the last stop word.**
A template added a brand suffix and truncated at 60 characters. 361 titles ended on "the" or "for".

- 03 The same paragraph on 121 pages with the numbers swapped.**
Exact-match dedupe never saw it. Google's spam policy calls it scaled content abuse.

- 04 An H2 with nothing under it.**
Twelve headings generated, nine filled. The other three rendered as a promise with no delivery.

- 05 An anchor that names one district and links to another.**
The reader clicks "Playa Norte overview" and lands in Playa Sur.

- 06 A page merged away six weeks ago, still advertised in llms.txt.**
Answer engines read the index instead of crawling. They cited a redirect.

- 07 "7.5% net yield" that nobody ever sourced.**
On 70 pages. Buyers who did the maths did not come back.

How to read this document. Pages 3 to 6 walk through the sixteen gates in four groups: mechanical, corpus debt, links and routing, trust. Page 7 shows how to run them. Page 8 shows what to do when they find more than you can fix.

Mechanical: minutes to fix

Deterministic checks. Fix these first; they take minutes and they hide the real findings behind noise.

01

Characters FAIL

FAILURE Currency signs in tables, emoji in headings, em dashes auto-corrected on every line.

```
bad-characters.mdx:8    currency sign ¢: spell it CRC
bad-characters.mdx:8    decorative pictograph 🏠
bad-characters.mdx:8    em or en dash: use a comma, a colon, parentheses or a hyphen in
                        ranges
```

FIX `gates --fix` rewrites signs to codes and removes pictographs. Dashes are left to a person: the right replacement depends on the sentence.

02

MDX compile FAIL

FAILURE An eleven-minute build fails on file 1,214 because of a quote inside a JSX string. Fix, push, wait, file 1,380.

```
bad-mdx.mdx    Expected a closing tag for '<Callout>' (10:1-10:22)
```

FIX Every file is compiled up front with the build's own parser. All failures at once, with the line.

03

Frontmatter FAIL

FAILURE Missing fields, cut titles, descriptions ending in an ellipsis, hero images on a stock host that changed its hotlink policy, related slugs pointing at pages that no longer exist.

```
bad-frontmatter.mdx    title length 17, expected 40 to 60
bad-frontmatter.mdx    description ends on a stop word (cut)
bad-frontmatter.mdx    relatedSlug "ghost-slug" does not exist
```

FIX Rewrite the field. For cut titles, drop the brand suffix rather than the promise.

04

Structure FAIL

FAILURE A commercial page that reads well and answers nothing: no answer at the top, one table, two H2s, no risks, four numbers in two thousand words. Nothing to quote, nothing to decide with.

```
bad-structure.mdx    missing answer-first block (Quick answer or TL;DR)
bad-structure.mdx    heading level skip H2 to H4 at line 12
bad-structure.mdx    draft or source marker left in text ([VERIFY], TODO, source
                        needed)
```

FIX Answer-first block, four H2s, six table rows, the FAQ count the collection needs, pros and cons, risks, scenarios, three numeric facts per 500 words, five internal links. Thresholds live in the config.

Corpus debt: where the work is

These three gates find what generated content leaves behind. They are the reason the package exists.

05

Duplication FAIL

FAILURE 121 project cards with the same closing paragraph and different numbers. The exact-match checker passed every one of them.

2 pages share: "the first 50 metres from the high tide line belong to the state and cannot be sold..."

2 pages share a template with swapped numbers: "median asking price per sqm in # is # crc, based on the registry..."

HOW Every paragraph of twelve words or more, compared exactly and with the digits masked, across pages and within a page. Same-shaped tables are structure, not prose, and are not counted. Disclaimers that must repeat go into a **boilerplate** list.

FIX One page keeps the paragraph, the others link to it. For templates with swapped numbers, write the sentence from the fact, not from the template.

06

Shared sentences FAIL

FAILURE The block check passed. The corpus still read as machine-written: one sentence, inside different paragraphs, on 454 pages.

3x sentence: "every lease longer than three years must be registered at the land registry within 30 days"

HOW Sentences of ten words or more, list items and table cells of eight words or more, normalised and counted across indexable pages. Three pages is a finding.

FIX Same as gate 05, one level down.

07

Hollow sections FAIL

FAILURE An outline with twelve H2s, nine written. Sections that open by restating their heading. Eleven FAQ answers that all say "Yes, in most cases. Contact our team."

hollow.mdx empty section "What the charge covers"

hollow.mdx section "How the charge is set" holds only shared filler

hollow.mdx numbered list or steps start above 1

hollow.mdx 1 table cell(s) holding only a comma

FIX Write the section or delete the heading. A heading that promises and delivers nothing costs more than no heading.

Links and routing: usually one config change

Five gates that fail quietly between deploys and are cheap to keep green once the pipeline generates the right files.

08 Internal links FAIL

FAILURE Two content waves merged on separate branches. Neither had a broken link. The merged site had forty.

```
/guides/does-not-exist/ (1 page(s), e.g. index.html)
```

HOW Every href, src and action in the built HTML resolved against the build and the redirects file. Without a build, source links are checked against the content index.

09 Link semantics WARN

FAILURE An anchor names one district, the href goes to another. A project name links to a generic guide instead of the project's page. One pillar linked four times on one screen.

```
bad-links.mdx  anchor names playa norte but link goes to playa sur: [Playa Norte
overview](/areas/playa-sur/)
bad-links.mdx  anchor is an entity name but links elsewhere: [Marina Towers]
(/guides/clean-guide/)
```

FIX Make the anchor say where the link goes. Entities link to their own page.

10 Meta WARN

FAILURE "Condo Guide 2025 vs 2026: Buyer Guide". The year twice, the format word twice, forty per cent of the title spent on nothing.

```
bad-meta.mdx  title repeats the year    ·    title repeats "guide"
```

13 Redirects FAIL

FAILURE A merged into B, then B into C. Search engines follow two hops grudgingly and three not at all. A page that still existed was redirected away.

```
cycle: /old-a -> /old-b -> /old-a
chain of 3: /x1 -> /x2 -> /x3 (point the first source at the final target)
live page /guides/bad-meta/ is redirected away
```

14 AI index FAIL

FAILURE llms.txt generated once. A month later: four pages merged away, one set to noindex, two collections added. Answer engines kept reading the old file.

```
index lists a page that does not exist: /guides/ghost-page/
index lists noindex page /guides/draft-page/
indexable page missing from the index: /guides/bad-structure/
```

FIX Generate the index from the corpus in the build. Run the gate after the build.

Trust: editorial work

The last three gates are about whether a person, a quality rater or an answer engine should believe the page.

11

Readability WARN

FAILURE A page that passed every structural gate and still read as generated: a dash every second line, four "Scenario A to D" blocks, sixty-word sentences.

```
bad-readability.mdx  5 sentences over 45 words
```

HOW Dashes per 500 words, template scenario blocks, corpus stamps left in text, sentences over 45 words, paragraphs over 170 words. Warns rather than fails: a legal page may legitimately run long.

12

Images FAIL

FAILURE A CDN account migrated. 2,300 hero images returned 404 for nine days. The pages still built.

```
bad-images.mdx  local image missing in public: /images/missing-hero.jpg
```

HOW Every image URL under `src/` gets a HEAD request, twelve in parallel with one retry. Local paths are checked against `public/`. Run it on a schedule: images break between deploys.

15

Unsourced claims FAIL

FAILURE "A coastal one-bedroom earns a 7.5% net yield with 82% occupancy." No source, no method, on 70 pages.

```
bad-claims.mdx  yield or ROI figure without a source: "7.5% net yield"
bad-claims.mdx  nightly rate without a source: "$140 per night"
```

HOW Yield, ROI, occupancy, nightly rate, income per period and appreciation figures in every paragraph. A paragraph passes when it names an allowed source, uses a sourcing phrase, or frames the figure as a worked example.

FIX Name the source in the same paragraph, turn the figure into an example with stated assumptions, or delete it. This is the gate that takes days, and the one that decides whether the site is believed.

16

Agent surface WARN

FAILURE A site published an agent card and markdown copies of its pages, then renamed a collection. The copies stopped being generated. Nobody noticed for two months: no human opens those URLs.

```
no .well-known/agent.json: an agent directory has nothing to read about this site
/guides/buying-costs/  page has no markdown rendition while 17 other page(s) do
```

HOW Two files an assistant looks for: the agent card at `/.well-known/agent.json` and a markdown rendition of each page. Both standards are early, so a missing file is a warning. A card that is broken JSON, or markdown that exists for some pages and not others, is a failure:

Three steps, no setup ceremony

- 1 Install.** Node 20 or newer, any folder of MDX or Markdown with frontmatter (Astro content collections work out of the box).
`npm i -D @operstack/gates`
- 2 Describe the site.** One JSON file at the project root. Or skip it: the tool detects `src/content/*` and applies default thresholds.
- 3 Run.** `npx gates`. Terminal table, Markdown report in `.gates/`, JSON for CI. Exit code 1 when something must not ship.

See every gate fire first. `npx`

`@operstack/gates --demo` runs the fictional Isla Verde site that ships with the package: one deliberate defect per gate and one page that passes all sixteen.

Flags. `--only 1,4,16`, `--fix`, `--offline`, `--fail-on warn`, `--json`.

```
{
  "siteUrl": "https://example.com",
  "contentDir": "src/content",
  "collections": {
    "guides": { "url": "/guides/",
      "minWords": 2000, "faq": 5,
      "commercial": true },
    "projects": { "url": "/projects/",
      "minWords": 400,
      "kind": "entity" }
  },
  "currency": [{ "sign": "฿", "code": "THB" }],
  "placeNames": ["bang tao", "kamala"],
  "boilerplate": ["^this card records"],
  "claims": { "allowedSources":
    ["central bank", "land registry"] }
}
```

```
# .github/workflows/gates.yml
- run: npm run build
- run: npx gates --fail-on warn
```

OperStack Gates

```
-----
OK      01 characters          corpus clean
OK      02 mdx-compile        1660 file(s) compile
FAIL    03 frontmatter        59 issue(s)
FAIL    05 duplication        9 shared block(s) on 64 page(s), 30 near-duplicate
FAIL    06 shared-sentences   27 sentence(s) and 4 fragment(s) on 3+ pages
WARN    09 link-semantics     205 anchor(s) disagree with their target
FAIL    13 redirects         4 redirect problem(s) in 268 rules
FAIL    14 ai-index          4 disagreement(s) between public/llms.txt and the site
FAIL    15 claims            156 unsourced figure(s)
-----
pass 3  warn 3  fail 9  skip 0          1,660 files, 12.9 s
```

A real run on a production corpus, abridged. The findings are the site's homework for the month.

Tell the model before the gate tells you. The package includes a rules file for Cursor and Claude Code (`rules/content-gates.md`). It states every rule the gates enforce, in the language a model follows, so the model stops producing what the gates reject.

When the gates find more than you can fix

The gates are free and stay free. They came out of a pipeline that builds and runs content sites; the rest of that pipeline is available in three sizes.

STEP	WHAT YOU GET	PRICE
Sixteen gates	The npm package, the rules file, this document, the fixture site. MIT licence, no account.	Free
Site Kit	The full pipeline the gates came from: Astro templates, page generator, mass indexing, AI index, image handling, a demo market, the gates wired in, quick start and documentation.	79 USD
SEO, AEO and GEO audit	We run the gates and eleven more checks on your site: search and answer-engine visibility, indexing, structure, trust. 18-page PDF with findings priced per fix. With Search Console access for the full data.	249 / 349 USD
Fix packages	Fix, Foundation, Growth: the findings from the audit applied to your repository or CMS as reviewed changes, measured at 30, 60 and 90 days.	from 600 USD

About OperStack

OperStack builds the tooling behind a pipeline that publishes to eleven content sites in several languages, with leads arriving from search and from ChatGPT, Perplexity and Copilot referrals. Every product is a piece of that pipeline, cleaned of client data and packaged so that one person can run it.

Where to find it

Package: `npx @operstack/gates`

Source and issues: github.com/oper-stack/gates

Site Kit, audit, documentation: oper-stack.com

Mail: hello@oper-stack.com

The thresholds in the config are opinions from one pipeline. Tell us which ones are wrong for your corpus; that is more useful than defending them.